

Particle Physics Experiment Lab.

Future Lepton Collider, KaeM, CMS, Medical Physics

June 04, 2025, Sehwook Lee

Research Opportunities

- Future Lepton Collider Experiments
  - Dual-Readout Calorimeter R&D
- CMS Experiment
  - Search for a new CP Violation source using top quark pair production
  - Search for ttHH production
  - Independent search using diphoton
- Tabletop Experiment
  - Search for low-mass, low-charge magnetic monopoles (KaeM):
    - R&D: Electroluminescent-TPC, Crystal calorimeter, Data Acquisition System
    - Magnetic bottle simulation, Background study, magnetic monopole interactions with materials
- Medical Physics
  - Digital X-ray diagnostic device for mammography